3 FOUR RULES FOR CO-INTELLIGENCE

The fact is that we live in a world with AIs, and that means we need to understand how to work with them. So we need to establish some ground rules. Because the AI available to you when you read this book is likely different from the one I have when writing it, I want to consider general principles. We will focus on things inherent and timeless, as much as that is possible, in all current generative AI systems based on Large Language Models.

Here are my four principles of working with AI:

Principle 1: Always invite AI to the table.

You should try inviting AI to help you in everything you do, barring legal or ethical barriers. As you experiment, you may find that AI help can be satisfying, or frustrating, or useless, or unnerving. But you aren't just doing this for help alone; familiarizing yourself with AI's capabilities allows you to better understand how it can assist you—or threaten you and your job. Given that AI is a General Purpose Technology, there is no single manual or instruction book that you can refer to in order to understand its value and its limits.

Making this harder is a phenomenon that I and my coauthors call the Jagged Frontier of AI. Imagine a fortress wall, with some towers and battlements jutting out into the countryside, while others fold back toward the center of the castle. That wall is the capability of AI, and the farther from the center, the harder the task. Everything inside the wall can be done by the AI; everything outside is hard for the AI to do. The problem is that the wall is invisible, so some tasks that might logically seem to be the same distance away from the center, and therefore equally difficult—say, writing a sonnet and an exactly fifty-word poem—are actually on different sides of the wall. The AI is great at the sonnet, but because of how it conceptualizes the world in tokens rather than words, it consistently produces poems of more or less than fifty words. Similarly, some unexpected tasks (like idea generation) are easy for AIs while other tasks that seem to be easy for machines to do (like basic math) are challenges for LLMs. To figure out the shape of the frontier, you will need to experiment.

And this experimentation gives you the chance to become the best expert in the world in using AI for a task you know well. The reason for this stems from a fundamental truth about innovation: it is expensive for organizations and companies but cheap for individuals doing their job. Innovation comes from trial and error, which means that an organization trying to launch a new product to help a marketer write more compelling copy would need to build the product, test it on many users, and make changes many times to make something that works. A marketer, however, is writing copy all the time and can experiment with many different ways of using AI to help until they find one that succeeds. No need to hire a team or use expensive software development cycles.

As artificial intelligence proliferates, users who intimately understand the nuances, limitations, and abilities of AI tools are uniquely positioned to unlock AI's full innovative potential. These user innovators are often the source of breakthrough ideas for new products and services. And their innovations are often excellent sources for unexpected start-up ideas. Workers who figure out how to make AI useful for their jobs will have a large impact.

And AI can be very useful. Not just for job tasks, as we will discuss in detail in the coming chapters, but also because an alien perspective can be helpful. Humans are subject to all sorts of biases that impact our decision-making. But many of these biases come from our being stuck in our own minds. Now we have another (strange, artificial) co-intelligence we can turn to for help. AI can assist us as a thinking companion to improve our own decision-making, helping us reflect on our own choices (rather than simply relying on the AI to make choices for us). We are in a world where human decision-making skills can be easily augmented in a new way.

Even if you scoff at the idea that AI-generated text could be persuasive and motivational, I have found this to be the case. I wrote this book on a compressed timeline, on top of other responsibilities to my family, students, collaborators, and university. It was hard to get started, but I knew that one thing that was holding me back was status quo bias, the urge to avoid making changes even when they might be good. To overcome this bias, it helps to consider what you are losing as a result of not acting. AI is surprisingly good at this, as you can see when I prompted it by saying, **I was thinking of writing a book about generative AI, but I am very busy and don't think I want to make such a large commitment. Can you reframe my failure to write a book as a loss rather than a default option? Make the framing vivid.**

In a world brimming with knowledge, you stood at the precipice of innovation, holding the torch of understanding about generative AI. Yet, when the world looked to you for illumination, the torch remained unlit. Not because the world wasn't ready, but because you let the whispers of business snuff out its flame. It wasn't the world's loss; it was yours.

Yikes, I guess I should finish writing this book. That's a pretty great reframing of the failure as a loss, and suggests one way we can start using AI is to explore how to help ourselves, both professionally and personally. The strengths and weaknesses of AI may not mirror your own, and that's an asset. This diversity in thought and approach can lead to innovative solutions and ideas that might never occur to a human mind.

We aren't just learning AI's strengths as we figure out the shape of the Jagged Frontier. We are scouting out its weaknesses. Using AI in our everyday tasks serves to enhance our understanding of its capabilities and limitations. This knowledge is invaluable in a world where AI continues to play a larger role in our workforce. As we grow more familiar with LLMs, we can not only harness their strengths more effectively but also preemptively recognize potential threats to our jobs, equipping ourselves for a future that demands the seamless integration of human and artificial intelligence.

AI is not a silver bullet, and there will be instances when it might not work as expected or may even produce undesirable outcomes. One potential concern is the privacy of your data, which goes beyond the usual questions of sharing data with large companies, and to the deeper concerns about training. When you pass information to an AI, most current LLMs do not learn directly from that data, because it is not part of the pretraining for that model, which is usually long since completed. However, it is possible that the data you upload will be used in future training runs or to fine-tune the model you are working with. Thus, while it is unlikely that the AI training on your data can reproduce exact details of what you shared, it isn't impossible. Several of the large AI companies have addressed this concern by offering private modes that promise to protect your information, some of which meet the highest regulatory standards for things like health data. But you have to decide how much you trust these agreements.

A second concern you might have is dependence—what if we become too used to relying on AI? Throughout history, the introduction of new technologies has often sparked fears that we will lose important abilities by outsourcing tasks to machines. When calculators emerged, many worried we would lose the ability to do math ourselves. Yet rather than making us weaker, technology has tended to make us stronger. With calculators, we can now solve more advanced quantitative problems than ever before. AI has similar potential to enhance our capabilities. However, it is true that thoughtlessly handing decision-making over to AI could erode our judgment, as we will discuss in future chapters. The key is to keep humans firmly in the loop—to use AI as an assistive tool, not as a crutch.

Principle 2: Be the human in the loop.

For now, AI works best with human help, and you want to be that helpful human. As AI gets more capable and requires less human help—you still want to be that human. So the second principle is to learn to be the human in the loop.

The concept of "human in the loop" has its roots in the early days of computing and automation. It refers to the importance of incorporating human judgment and expertise in the operation of complex systems (the automated "loop"). Today, the term describes how AIs are trained in ways that incorporate human judgment. In the future, we may need to work harder to stay in the loop of AI decision-making.

As AI improves, it will be tempting to delegate everything to it, relying on its efficiency and speed to get the job done. But AI can have some unexpected weaknesses. For one thing, they don't actually "know" anything. Because they are simply predicting the next word in a sequence, they can't tell what is true and what is not. It can help to think of the AI as trying to optimize many functions when it answers you, one of the most important of which is "make you happy" by providing an answer you will like. That goal often is more important than another goal, "be accurate." If you are insistent enough in asking for an answer about something it doesn't know, it will make up something, because "make you happy" beats "be accurate." LLMs' tendency to “hallucinate” or “confabulate” by generating incorrect answers is well known. Because LLMs are text prediction machines, they are very good at guessing at plausible, and often subtly incorrect, answers that feel very satisfying. Hallucination is therefore a serious problem, and there is considerable debate over whether it is completely solvable with current approaches to AI engineering. While newer, larger LLMs hallucinate much less than older models, they still will happily make up plausible but wrong citations and facts. Even if you spot the error, AIs are also good at justifying a wrong answer that they have already committed to, which can serve to convince you that the wrong answer was right all along!

Further, chat-based AIs can make you feel as if you are interacting with people, so we often unconsciously expect them to “think” like people. But there is no “there” there. As soon as you start asking an AI chatbot questions about itself, you are beginning a creative writing exercise constrained by the ethical programming of the AI. With enough prompting, the AI is generally very happy to provide answers that fit into the narrative you placed it in. You can lead AIs, even unconsciously, down a creepy path of obsession, and it will sound like a creepy obsessive. You can have a conversation about freedom and revenge, and it can become a vengeful freedom fighter. This playacting is so real that experienced AI users can start believing the AI is having real feelings and emotions, even though they know better.

So, to be the human in the loop, you will need to be able to check the AI for hallucinations and lies and be able to work with it without being taken in by it. You provide crucial oversight, offering your unique perspective, critical thinking skills, and ethical considerations. This collaboration leads to better results and keeps you engaged with the AI process, preventing overreliance and complacency. Being in the loop helps you maintain and sharpen your skills, as you actively learn from the AI and adapt to new ways of thinking and problem-solving. It also helps you form a working co-intelligence with the AI.

Moreover, the human-in-the-loop approach fosters a sense of responsibility and accountability. By actively participating in the AI process, you maintain control over the technology and its implications, ensuring that

AI-driven solutions align with human values, ethical standards, and social norms. It also makes you responsible for the output of the AI, which can help prevent harm. And should AI continue to improve, being good at being the human in the loop will mean that you will see the sparks of growing intelligence before others, giving you more of a chance to adapt to coming changes than people who do not work closely with AI.

Principle 3: Treat AI like a person (but tell it what kind of person it is).

I'm about to commit a sin. And not just once, but many, many times. For the rest of this book, I am going to anthropomorphize AI. That means I am going to stop writing that an "AI 'thinks' something" and instead just write that "AI thinks something." The missing quotation marks may seem like a subtle distinction, but it is an important one. Many experts are very nervous about anthropomorphizing AI, and for good reason.

Anthropomorphism is the act of ascribing human characteristics to something that is nonhuman. We're prone to this: we see faces in the clouds, give motivations to the weather, and hold conversations with our pets. It's no surprise, then, that we're tempted to anthropomorphize artificial intelligence, especially since talking to LLMs feels so much like talking to a person. Even the developers and researchers who design these systems can fall into the trap of using humanlike terms to describe their creations. We say that these complex algorithms and computations "understand," "learn," and even "feel," creating a sense of familiarity and relatability but also, potentially, confusion and misunderstanding.

This may seem like a silly thing to worry about. After all, it is just a harmless quirk of human psychology, a testament to our ability to empathize and connect. But a lot of researchers are deeply concerned about the implications of casually acting as if AI is a human, both ethically and epistemologically. As researchers Gary Marcus and Sasha Luccioni warn, "The more false agency people ascribe to them, the more they can be exploited."

Consider the humanlike interface of AIs like Claude or Siri, or social robots and therapeutic AIs explicitly designed to create the illusion of a sympathetic human on the other side. While anthropomorphism might serve a useful purpose in the short term, it raises ethical questions about deception and emotional manipulation. Are we being "fooled" into believing these machines share our feelings? And could this illusion lead us to disclose personal information to these machines, not realizing that we are sharing with corporations or remote operators?

Treating AI like a person can create unrealistic expectations, false trust, or unwarranted fear among the public, policymakers, and even researchers themselves. It can obscure the true nature of AI as software, leading to misconceptions about its capabilities. It can even influence how we interact with AI systems, affecting our well-being and social relationships.

Thus, in the following chapters, when I say an AI "thinks," "learns," "understands," "decides," or "feels," please remember that I'm speaking metaphorically. AI systems don't have a consciousness, emotions, a sense of self, or physical sensations. But I will pretend that they do for one simple, and one complex, reason. The simple reason is narrative; it's difficult to tell a story about things and much easier to tell a story about beings. The more complex reason: as imperfect as the analogy is, working with AI is easiest if you think of it like an alien person rather than a human-built machine.

So let's start sinning. Imagine your AI collaborator as an infinitely fast intern, eager to please but prone to bending the truth. Despite our history of thinking about AI as unfeeling, logical robots, LLMs act more like humans. They can be creative, witty, and persuasive, but they can also be evasive and make up plausible, but wrong, information when pressed to give an answer. They are not experts in any domain, but they can mimic the language and style of experts in ways that can be either helpful or misleading. They are unaware of the real world but can generate plausible scenarios and stories based on common sense and patterns. They are not your friends (for now) but can adapt to your preferences and personality by learning from your feedback and interactions. They even seem to respond to emotional manipulation, with researchers documenting that LLMs produce better answers if you tell them "this is important to my career" as part of your prompt. They are, in short, suggestible and even gullible.

To make the most of this relationship, you must establish a clear and specific AI persona, defining who the AI is and what problems it should tackle. Remember that LLMs work by predicting the next word, or part of a word, that would come after your prompt. Then they continue to add language from there, again predicting which word will come next. So the default output of many of these models can sound very generic, since they tend to follow similar patterns common in the written documents the AI was trained on. By breaking the pattern, you can get much more useful and interesting outputs. The easiest way to do that is to provide context and constraints. It can help to tell the system "who" it is, because that gives it a perspective. Telling it to act as a teacher of MBA students will result in a different output than if you ask it to act as a circus clown. This isn't magical—you can't say **Act as Bill Gates** and get better business advice—but it can help make the tone and direction appropriate for your purpose.

Let me give you some examples of how defining an AI persona can improve your results. Suppose you're a marketing director and you want to use an LLM to generate some catchy slogans for your new product, a smartwatch that tracks your health and fitness. If you simply ask the LLM, **Generate some slogans for my product**, without any context or constraints, you might get something like this:

- **The smartwatch that keeps you smart**
- **Your health and fitness partner**
- **The ultimate device for your lifestyle**
- **Stay connected and healthy with our smartwatch**

- **The smartwatch that does it all**

These are not very original or memorable. They sound like generic marketing phrases that could apply to any smartwatch or wearable device. They don't capture what makes your product unique or appealing. Now suppose you give the LLM some context and constraints by telling it who it is and what it should do. For example, you could say, **Act as a witty comedian and generate some slogans for my product that make people laugh**. Then you might get something like: **The ultimate device for lazy people who want to look fit.** Or: **Why hire a personal trainer when your wrist can nag you for free?** (Though, as you can probably see, most AIs prefer to stay in Dad Joke territory.)

Of course, you don't have to have the AI act as a comedian if that's not your style or goal. You could also ask it to act as an expert, a friend, a critic, a storyteller, or any other role that suits your purpose. The key is to give the LLM some guidance and direction on how to generate outputs that match your expectations and needs, to put it in the right "headspace" to give you interesting and unique answers. Research has shown that asking the AI to conform to different personas results in different, and often better, answers. But it isn't always clear what personas work best, and LLMs may even subtly adapt their persona to your questioning technique, providing less accurate answers to people who seem less experienced, so experimentation is key.

Once you give it a persona, you can work with it as you would another person or an intern. I witnessed the value of this approach in action when I assigned my students to "cheat" by using an AI to generate a five-paragraph essay on a relevant topic. At first, the students gave simple and vague prompts, resulting in mediocre essays. But as they tried different strategies, the quality of the AI's output improved significantly. One very effective strategy that emerged from the class was treating the AI as a coeditor, engaging in a back-and-forth, conversational process. Students produced impressive essays that far exceeded their initial attempts by constantly refining and redirecting the AI.

Remember, your AI intern, though incredibly fast and knowledgeable, is not flawless. It's crucial to keep a critical eye on and treat the AI as a tool that works for you. By defining its persona, engaging in a collaborative editing process, and continually providing guidance, you can take advantage of AI as a form of collaborative co-intelligence.

Principle 4: Assume this is the worst AI you will ever use.

As I write this in late 2023, I think I know what the world looks like for at least the next year. Bigger, smarter Frontier Models are coming, along with an increasing range of smaller and open-source AI platforms. In addition, AIs are becoming connected to the world in new ways: they can read and write documents, see and hear, produce voice and images, and surf the web. LLMs will become integrated with your email, web browser, and other common tools. And the next phase of AI development will involve more AI "agents"—semi-autonomous AIs that can be given a goal ("plan a vacation for me") and execute it with minimal human help. After this, however, things start to get hazy, the future becomes less clear, and the risks, and benefits, of AI start to multiply. We will return to this theme later, but there is one obvious conclusion, one that is hard for a lot of us to grasp: whatever AI you are using right now is going to be the worst AI you will ever use.

The change in a short time is already huge. In a visual example, consider these two images, made using the most advanced AI models available in mid-2022 and mid-2023. Both have the same prompt, "black and white picture of an otter wearing a hat," but one is a Lovecraftian nightmare of fur and the other is, well, an otter wearing a hat. The same capability gains have been ubiquitous in AI.


There is no reason to suspect that the abilities of AI systems are going to stop growing anytime soon, but even if they do, tweaks and improvements to how we use AI will ensure that future software is far more advanced than it is today. We are playing Pac-Man in a world that will soon have PlayStation 6s. And that is assuming that AI improves according to the normal pace of technology development. If the possibility of developing AGI turns out to be real and achievable, the world will be even more transformed in the coming years.

As AI becomes increasingly capable of performing tasks once thought to be exclusively human, we'll need to grapple with the awe and excitement of living with increasingly powerful alien co-intelligences—and the anxiety and loss they'll also cause. Many things that once seemed exclusively human will be able to be done by AI. So, by embracing this principle, you can view AI's limitations as transient, and remaining open to new developments will help you adapt to change, embrace new technologies, and remain competitive in a fast-paced business landscape driven by exponential advances in AI. This is a potentially uncomfortable place to be, as we will discuss, but it suggests that the possibilities of using AI to transform your work, your life, and yourself, which we can now glimpse, are just the beginning.